

Weekly Notice 5

Exercises, week 5:

1. Exercise 41(1). Use Proposition 6.12.
2. Exercise 44.
3. Exercise 48.
4. Exercise 51. Based on the material covered in the lecture Monday, February 23.
5. Exercise 52.
6. Let S be countable and $\mathbf{P} = \{p_{ij}\}_{i,j \in S}$ denote a transition matrix. Consider the usual measures $\mathbb{P}_i$ and $\mathbb{P}_\mu$ on the canonical path space. Using Equation (4.28) in the lecture notes show that, regardless of the initial distribution μ ,

$$\mathbb{P}_\mu = \sum_{i \in S} \mu(\{i\}) \mathbb{P}_i.$$

That is,

$$\mathbb{P}_\mu(A) = \sum_{i \in S} \mu(\{i\}) \mathbb{P}_i(A) \quad \text{for } A \in \mathcal{B}(S)^{\mathbb{N}_0}.$$

7. Exercise 67.